



The Open University

Undergraduate Computing
M359 Relational databases: theory and practice

M359 Glossary

1NF	see first normal form	
2NF	see second normal form	
3NF	see third normal form	
absolute path		A path in a tree structure that starts at the root node.
acceptance testing		Demonstrating to the client that the database developed satisfies the information requirements of the enterprise.
access control		The DBMS facility that ensures that only authorised users gain access to the DBMS.
adaptive maintenance		Concerns restructuring the database to meet new and changing requirements.
aggregate function		A function that acts upon multiple rows in a table to produce a single value result.
aggregate navigator		A component of the data warehouse architecture that sits between the OLAP server and DBMS(s) that manage the warehouse database. The role of the aggregate navigator is to ensure the effective and transparent use of materialized views.
aggregate rows		In the Logical Processing Model (see Block 3) the rows of the table that results from the processing of a SELECT clause after a GROUP BY clause.
aggregation		The combination of several facts into a single measure, performed during a roll-up operation. Also known as summarisation.
alias (relational)		Acts as a placeholder or temporary name for another relational algebra expression. <i>TemporaryName alias ComplicatedExpression</i>
alias (SQL)		A temporary alternative name for a particular table within a query.
alternate key		A candidate key which has not been chosen to be a primary key.
amendment anomaly		An anomalous situation in which replication of data makes it difficult to amend occurrences of single-valued facts.
analytical processing		Processing that involves the analysis of data, rather than simple transaction recording. Usually not interactive, although with increasingly powerful computer systems interactive analytical systems are becoming possible.
application	see user process	Usually an abbreviated form of user application.
application programming interface (API)		The means by which an application program interacts with a set of system services, such as an operating system or a DBMS; also known as application program interface.

arguments (of a function or procedure)		The values passed to a function or procedure.
assertion		A formal statement that is expected to be true. E.g. Stating that there is an entity instance for Student(S01, Thompson,...) is an assertion that there is a student in the University with the identifier S01 and the name Thompson.
assumptions		In the conceptual model these are a list of the decisions or questions made by the modeler that require some form of review by the client. Assumptions recognise that not every aspect of the domain of discourse will be exactly defined prior to modeling and allow the model to be constructed with the points needing clarification being noted.
attribute (of a relation)		Consists of a name and a domain from which values of the attribute are drawn.
attribute (of an entity type)		A property or characteristic of an entity type. E.g. The Student has a Name.
attribute (of an XML element)		A property of an XML element that can have a single value.
attribute value		The value of a property or characteristic of an entity. E.g. That student has the Name "Thompson".
augmentation		This is a property of functional dependencies which states that if A, B and C are sets of attributes of a relation R such that $A \mapsto B$, then $A, C \mapsto B, C$.
authentication		Ensuring that the user of an authorisation identifier is entitled to use that identifier. A password is often used as the authentication mechanism.
authorisation identifier		A name provided to the database that allows a user to connect to a database. Used by the DBMS to determine what data in the database a user may query or modify.
backup copy		A copy of the data held in a database made at some point in time. It is used if the original data becomes corrupt or damaged as a result of a system failure or other error.
backward compatibility		Where a version of the system can use data or programs written for earlier versions of the system.
base table		A table of data in the database that actually contains data values (in contrast to a view, in which data is described by a query).
BCNF	see Boyce-Codd normal form	
binary operator		A binary operator is an operator having two operands.
body (of a relation)		The body of a relation consists of the set of its tuples.
body (of a routine)		The statement or compound statement that is executed when a procedure or function is called.

Boolean expression		An expression which evaluates to either true or false.
Boyce-Codd normal form		A relation is in Boyce–Codd normal form (BCNF) if and only if each irreducible determinant of a non-trivial functional dependency is a candidate key.
built-in functions		Functions defined as part of Standard SQL.
bulk load		Entering data directly into a database without using interactive user-interfaces. Usually bulk loaded data has been exported from another database. Bulk loading usually implies the suspension of consistency and correctness checking until the entire data loading is complete.
business application logic		The business programming that is designed to satisfy the business requirements for an information system. Usually coded in the user processes, but can be distributed in a client-server system. For example, calculating a billing total based on a work ticket.
candidate key	see also primary key, alternate key	A set of attributes K is a candidate key for a relation R if and only if it possesses the following two properties: (i) Uniqueness. It is illegal for R to contain two distinct tuples with the same value for K . (ii) Minimality. No proper subset of K has the uniqueness property. A candidate key may be a primary or alternate key.
candidate key constraint		A candidate key of a relation imposes a constraint on the values of the tuples forming the body of that relation in that two distinct tuples cannot have the same values for the candidate key.
cardinality (of a relation)		The number of tuples in a relation.
Cartesian product (of tables)		The Cartesian product of two tables is a table which consists of all possible pairs of rows from the original two tables.
cascade effect		A means of ensuring referential integrity by deleting all tuples that reference a deleted tuple.
cast		A command to convert an item of data to a different type.
character data		the content of an XML document that is not markup.
check constraint		A condition defined on values in a table which must not evaluate to FALSE (so must evaluate to TRUE or UNKNOWN).
child node		A node in a data structure connected to one other node at a higher level (its parent node).
classical logic		(Broadly speaking) Logical systems in which every proposition is either TRUE or FALSE.
clause		A single line of an SQL statement, usually specified by the associated keyword (for example ‘the WHERE clause’).

client		The process in a client-server system that communicates a request to the server processes.
client application	see user process	
client multiserver		A client process that can access several database servers, but must do so by connecting to one server at a time.
client-server system		A computer architecture in which the various components are separate processes that can communicate by some mechanism. Frequently used to refer to components distributed over networked computers, but client-server systems can exist within a single computer.
closed (of operators)		An operator is closed over a set if, when it is applied to elements in the set, the result yielded is also an element of the set. For example, relational operators take relations as operands and yield a relation.
closure property		The property of a mathematical system that says that any operation on the base types always produces an object of the base types. The relational model has the closure property; any relational algebra operation applied to a relation will produce another relation as a result. Consequently, operations can be composed without ever producing a result that is not in the relational model.
column		A set of values in a table, all of which are the same type. A column contains one value for each row in the table.
column constraint		A constraint defined on a particular column of a table.
combining (of functional dependencies)		This is a property of functional dependencies, which says that if $S \mapsto A$ and $S \mapsto B$ then $S \mapsto A, B$ The converse also holds. That is, if $S \mapsto A, B$ then $S \mapsto A$ and $S \mapsto B$.
commit		In transaction management a commit instruction indicates to the DBMS that a transaction is complete and the changes requested can be made permanent in the database.
communication costs		The costs involved in transferring and presenting data in appropriate formats for authorised end users of an information system. This includes the costs of maintaining data (e.g. archiving) for long term use.
commutative		A function, $f(x,y)$, is commutative iff $f(x,y) = f(y,x)$ for all values on which f is defined. That is, a function is commutative if it gives the same answer whatever order its arguments appear. (Note iff means ‘if and only if’.)
comparison operators	see conditional expressions	Any of the operators $=, <, >, <=, >=$ and $<>$ (not equal to). The result of a comparison can be TRUE, FALSE or UNKNOWN.
composite query		A query containing more than one query specification.
compound statements		A number of statements that are collected together and executed as a single statement.

concept hierarchy		A description of how a dimension may be partitioned into different levels of granularity.
conceptual data model		An abstract model of data, its key properties and the associations that hold between data. In this course we use entity relationship models and relational representations for the conceptual data models.
conceptual schema	see conceptual data model	
concurrency management		The features of a computer system that permit several users or activities to occur together (notionally simultaneously) without the effects of one user or activity interfering with the other users or activities.
concurrent transactions		Transactions written by different users which are able to execute at the same time.
conditional expressions	see comparison operators	Expressions that evaluate to TRUE, FALSE or UNKNOWN.
conformance		An SQL implementation conforms to the SQL:2003 Standard if it implements core SQL.
connection management		Software components responsible for the connection between client and server components of a system. In a DBMS environment, this includes connecting users to specific databases.
constraint		A condition or conditions on data and its processing that prevent meaningless data being recorded in a database, and prevents meaningless operations being performed on the data.
constraint (relational representation)		Relational algebra expressions which, for a specific relation, prevent certain tuples from appearing in that relation.
constraint (conceptual data model)		In a conceptual data model, a list of the restrictions on the values of attributes, instances of entity types or occurrences of relationships that the model must satisfy in order for the data represented by the model to be considered accurate and consistent.
constraint (SQL)		A description of restrictions on the values that can appear in a table.
context node (XML)		The node that is currently being processed.
context path (XML)		A path in a tree starting from the context node.
control statement		A statement that determines the order in which the statements in a program are executed.
co-operative source (relating to information source)		Databases that support triggers, so that notifications of changes of interest can be recorded automatically.

core SQL		A set of features used by most dialects of SQL. An implementation which contains the features of Core SQL can claim conformance with the SQL:2003 Standard.
corporate information strategy		A business strategy that includes both business plans and guidelines for the information-based activities. Usually comprising three main areas: information systems, information technology and information management.
correlated subquery		A subquery that contains an outer reference.
crow's foot		The descriptive terms for the diagrammatic representation of the :m end of a relationship line in an entity-relationship diagram.
CRUD methods		Methods to create (i.e. insert), retrieve, update and delete rows of a table.
cube (relating to multi dimensional data model)		An operation takes an n-dimensional view of data and produces the other $2^n - 1$ different possible views, or aggregations, of the same data.
current connection		The active connection to a computer system.
current row (in SQL)		The row of a result set referenced by the current state of a cursor.
cursor		An SQL structure that identifies a particular row in a table.
cursor table		A derived table, which is used to fetch values into a cursor.
custom-built applications		A user process written specifically for a given set of requirements. Contrast with a general-purpose application, such as a spreadsheet, that can be adapted for several purposes.
data		Recordable facts about the world.
data cleaning/ cleansing		Dealing with the detection and removal of errors and inconsistencies in the data from a single information source in order to improve the quality of data. Also known as data scrubbing.
data control language		A programming language specifically intended to manage data access controls, permissions and constraints on database content. Compare with data manipulation language and data definition language.
data controller		A data controller is a person who determines the purpose for which, and the manner in which, any personal data are, or are to be, processed.
data definition		The description of the data used to instruct the DBMS to create the appropriate data storage and descriptions for the data and its constraints.
data definition language		A programming language specifically intended to manage the description of data and constraints in a database. Compare with data control language and data manipulation language.

data dependence		A consequence of file-based storage where the file-based processing is closely tied to the physical, fixed, structure of the data stored in the files.
data dictionary		A collection of catalogues that are intended to maintain descriptions of data and its properties (sometimes refers to the software used to manage the catalogues). Generally part of a software engineering environment, used specifically in a database context as a repository for descriptive text associated with requirements and modeling documents. Occasionally used as a synonym for the Systems Catalogue of a DBMS.
data independence		The separation of the physical storage details from the application processes using the data. See logical interface.
data island syndrome		Within an organisation the management strategy that permits individuals or groups to develop their own data management strategies, databases and data acquisition policies. This can result in the organisation losing track of the data it holds.
data manipulation		A query that involves changing or processing data held in the database. Generating a count of rows in a table, or updating values held in a row of a table are both examples of data manipulation.
data manipulation language		A programming language specifically intended to manage the querying of data. Compare with data control language and data definition language.
data mart		A data warehouse whose scope is confined to specific selected subject(s) of interest. See also data warehouse, enterprise data warehouse, distributed data warehouse.
data mining		The process of finding significant, previously unknown, and potentially valuable knowledge hidden in data. (typically identifying trends and patterns in data). Data mining provides the ability to automate the identification and extraction of information from an organisation's data warehouses, which can be used to make informed business decisions.
data requirements		The part of the information requirements that describe the data, and the relationships between the data, that need to be recorded by the database.
data scrubbing	see data cleaning	
data subject		The Data Protection Act, 1998 defined a Data Subject as an individual who is the subject of personal data.
data tagging		Adding textual labels to data items, usually to attempt to capture the semantics and structure of the data being labeled. E.g. <surname><Thompson><\surname>

data type		The set of values that an item of in a table may take. See also predefined data types and user-defined data type.
data warehouse		The database component of a decision support system. A data warehouse is a repository of consolidated historical data with a logical structure that enables decision makers to view and analyse the data in different, and multiple, dimensions. See also distributed data warehouse, virtual data warehouse, physical data warehouse
data warehousing		The process of building, managing and using data warehouses to support analytical data processing and business planning.
database		A collection of structured, persistent shared data, managed by a database management system, and having a logical interface.
database approach		A method of organizing and accessing data that is not dependent on knowing the fixed physical structure of the files in which the data is stored.
database development		The collection of tasks associated with those stages a database goes through during the building and installation of a new database that satisfies the information requirements of an enterprise.
database engine		The process (or processes) in a database management system that directly accesses and manipulates data.
database life cycle		The stages a database goes through, from the time the need for a database is established, until it is withdrawn from use.
database maintenance		The tasks that ensure the database continues to satisfy those requirements after it has been installed, and adapting the database to meet new and changing requirements.
database management system (DBMS)		A software system that is responsible for all interactions with data held in a database.
database procedure	see procedure	
database schema		The logical schema of a database.
database server	see database management system	Used to refer to a DBMS, a view that sees the DBMS as a server accepting data processing requests from client applications and returning the results when the processing has been performed.
database tool		A general-purpose software package that can be used with many different databases (one at a time), by any authorized user, to access any data available to that user within the database. Intended to offer flexible ad-hoc access to data.
DBMS system schema		Database tables that store data related to the DBMS and the databases, users, security requirements and so on.

decision making		The process of choosing among alternative courses of action for the purpose of attaining a goal or goals. Managerial decision making embodies the whole process of management: planning, organising and controlling. Planning, for example, involves a series of decisions: What should be done? When? How? Where? By whom?
decision support systems		Computer-based systems that incorporate data warehouses and employ data mining techniques to facilitate and improve strategic decision making by providing decision makers with relevant information
declaration (of relational heading)		This is of the form: relation <NameOfRelation> <NameOfAttribute1>: <DomainOfAttribute1> <NameOfAttribute2>: <DomainOfAttribute2> primary key <primary key> followed by foreign key declarations and constraint declarations where relevant.
default effect		A means of ensuring referential integrity by setting the value of a foreign key to a default when the referenced tuple is deleted.
default values (SQL)		A value defined for a column that is used if that column's value is not explicitly provided for a particular row.
degree (of a relationship)		A property of a relationship that indicates how two or more entity occurrences are related to each other. Degrees of interest are one-to-one, one-to-many and many-to-many.
degree (of a relation)		The number of attributes in the relation.
deletion anomaly		A situation in which it is impossible to delete an occurrence of a single-valued fact without deleting occurrences of other single-valued facts.
denormalisation		The process of merging tables to speed up the retrieval process by deliberately introducing redundancy into the data. The process can result in the formation of unnormalized tables.
dependent rows		A row that references a row in another table.
derivable properties		Properties where the associated data values can be obtained from the data values of one or more other properties. Data values that will be derived (secondary data) from other data collected and/or generated by an enterprise (primary data) are not usually included in an entity-relationship model as their inclusion results in redundancy in the model because of the duplication of the information. Such duplication is a source of inconsistency and should be avoided.

derived table		A table that is built using data in base tables. A derived table may be returned by a subquery or as the definition of a view.
design decisions		Choices made during the database design task on how to represent a entity-relationship model as a database schema. These choices arise because an entity-relationship model describes <i>what</i> the data requirements are, not <i>how</i> they should be implemented.
determinant (of a functional dependency)		This is the left-hand-side of a functional dependency. That is, given a functional dependency $S \mapsto A$, S is the determinant.
development lifecycle		A series of structured processes and activities involved in the development of a system, or product. Usually including analysis, design, implementation, testing and maintenance.
dialect (SQL)		An implementation of SQL.
difference (a relational operator)		Given union-compatible relations A and B , A difference B is the relation with heading the same as A and B and with body the set of all those tuples which are in the body of A but not in the body of B .
dimension table		A table in a ROLAP system that describes values along one dimension.
dimensions (of multi dimensional data model)		The different perspectives by which facts are viewed. Provide the context for facts about data. Dimensions are used for two purposes: <i>selection</i> of data and <i>grouping</i> of data at the desired level of detail. Each fact depends on a set of dimensions. Each dimension is defined over a specific domain.
dimension-value pairs		A specification of a value along a certain dimension. A set of dimension-value pairs references a cell in a multidimensional view of data.
direct access		Data items can be read directly (as opposed to sequentially), usually by using an index.
dirty read		An inconsistency that can arise during concurrent transactions. A user reads another user's uncommitted data that is subsequently rolled back, or changed before being committed. So the first user has read data that was never committed.
distributed data warehouse		An enterprise data warehouse that is formed from several independent data marts.
distributed database management system, DDBMS		A DBMS that allows user to access distributed data as if that data was held locally. The user does not see the distribution of data, all management of which is controlled by the DDBMS.
distributed system		A system in which the various facilities are formed from components executing on a network of computers.
distribution schema		A description of how the data in the global schema is physically distributed among the network of databases forming the distributed system.

divide		The general form of a divide expression is: divide <Relation1> by <Relation2> Here, the attributes of <i>Relation2</i> must be a subset of the attributes of <i>Relation1</i> (that is, each attribute of <i>Relation2</i> has the same name and is defined over the same domain as an attribute of <i>Relation1</i>). The divide expression evaluates to a relation having all those attributes which are in <i>Relation1</i> but not in <i>Relation2</i>. A tuple in this relation is of the form <a1, a2, ... > where <a1, a2, ... b1, b2...> appears in <i>Relation1</i> for all tuples <b1, b2...> of <i>Relation2</i>.
document declaration		The first markup in a document prolog, which specifies the version of XML being used and optionally its encoding.
Document Object Model		A tree structure interface to a parsed XML document, known as DOM.
document prolog		Markup that can occur before the root element of an XML document.
document type declaration		Markup in a document prolog that specifies, by inclusion or reference, the DTD (document type definition) to be used for validating an XML document.
Document Type Definition (DTD)		A formal description of the structure of an XML document expressed according to the XML specification, usually referred to as a DTD.
domain (relational model)		A named set of values from which one or more attributes draw their actual values.
domain (SQL)		A set of valid values that can be stored in the columns of a base table, defined in terms of data types.
domain of discourse		The commonly agreed definitions and descriptions that apply to the things within the scope of a database about which a client, developer and end-users will need to agree in order that there is no confusion or ambiguity in the meaning of the data or how it is manipulated.
drill down (relating to multi dimensional data model)		An operation is the reverse of the roll-up operation. The result of the operation is to move to a more detailed view of the data along one or more dimensions. The drill-down operation performs a distribution on a multidimensional view of data, either by descending a concept hierarchy for a dimension, or by increasing the number of dimensions.
drill up (relating to multi dimensional data model)	see roll up	
dynamic constraints		Constraints on the possible ways that a database may change.
dynamic SQL		SQL statements that are constructed during the execution of an application program.

eager approach (of data warehousing)		Information from each source is extracted, translated and cleaned as appropriate, merged with the relevant information from the other sources, and stored in a centralized repository. When a query is posed, it is evaluated directly at the repository, without needing to access the original information sources.
edge table		A table that can represent any XML document by recording the link from each node of the document to its parent with associated character data.
element		A component of an XML document that contains other elements or character data.
embedded SQL		SQL statements that occur within an application program.
empty element		An element with no content, denoted by a single empty-element tag.
encapsulation		A property of a data representation that hides the internal structure of the data from direct access, forcing use of specific procedures (methods, functions) to manipulate the data. The external appearance of the data items can be different from the internal representation. Commonly considered an object-oriented strength, but also present in abstract data types and other programming techniques.
end-tag		A label denoting the end of an element.
end-users		Those users of an information system who are using the system for their own purposes. Contrasted with, for example, system administrators.
enterprise data warehouse	see data warehouse, data mart	A data warehouse that covers all subjects of interest to an organization.
entity (conceptual model)	see entity occurrence	A common abbreviation for entity occurrence. (Warning, it can also be used, in some texts, as an abbreviation of entity type.)
entity (for XML)		A name that is used as a reference to a value in an XML document.
entity integrity rule		The rule that no column of a table that forms part of that table's primary key may contain NULL.
entity occurrence		An object of interest about which we want to record data. (An instance of an entity type).
entity sub-type		A concrete entity type that has occurrences that define the properties specific to that particular entity subtype. For a given supertype, there must be at least two subtypes.
entity super-type		An abstract entity type that has no occurrences and serves solely to define the properties common to all of their subtypes. The properties of a supertype are inherited by all of its subtypes.

entity type		A collection of objects of interest about which we want to record data. E.g. Student.
entry level SQL		A subset of SQL:1992, selected to ensure compatibility between generations of SQL.
erroneous data		data that may be classified as <i>incomplete</i> , <i>noisy</i> or <i>inconsistent</i> . <i>incomplete</i> : missing data, lacking attribute values or certain attributes of interest; <i>noisy</i> : data values that deviate from the expected; <i>inconsistent</i> : data that violates domain, uniqueness, referential or general constraints.
exclusive OR		In two valued logic, if OR is considered exclusive (usually written XOR), then the expression $A \text{ XOR } B$ is TRUE if A is TRUE or B is TRUE, but FALSE if both A and B are TRUE.
exclusive relationships		A pair, or more, of relationships whose occurrences cannot involve the same entity instance more than once. Recorded in our modeling by additional constraints statements.
execution costs		The costs involved in the day-to-day running of an information system.
extending determinants		This is a property of functional dependencies which states that if A , B and C are sets of attributes of a relation R , such that $A \twoheadrightarrow C$ and A is a subset of B , then $B \twoheadrightarrow C$ It follows that if $B \twoheadrightarrow C$ does not hold and A is a subset of B , then $A \twoheadrightarrow C$ does not hold.
extension		Functionality in an implementation of SQL that is not part of the Standard.
external schema		An external schema describes the data required by one or more application processes. A part of the three-schema architecture, there can be many external schema defined for a database.
extraction component (of data warehouse architecture)		The component copies operational data from the various information sources, translates it, and passes it to the integrator. Comprises the set of wrappers and monitors for a data warehouse used to extract data from the information sources.
fact constellation	see galaxy schema	
fact table		The table in a ROLAP system that contains unaggregated (raw) facts.
facts (of multi dimensional data model)		Numerical measures of the subject area of interest, typically sales. Each fact depends on a set of dimensions, which provides a context for the measure.
FD	see functional dependency	

file-based storage		A method of organizing and accessing data in fixed format files, where file names and structural information are required to locate and process data held in the files.
final table		The table that is returned as the result of a query, after all of the logical processing.
First normal form		A relation is in first normal form (1NF) if and only if it has no duplicate tuples and in each tuple, each value of every attribute is unique.
first-cut design		A design for a database that satisfies the conceptual data model requirements for the correctness and consistency of the data. The first cut design might lead to an efficient implementation on specific hardware or software.
flexing (design)		The collection of tasks associated with those stages a database goes through during the building and installation of a new database that satisfies the information requirements of an enterprise.
foreign key		A foreign key is an attribute (or combination of attributes) in one relation, R_2 , whose value in each tuple of R_2 appears as the value of a given candidate key (typically the primary key) of some relation, R_1 (where R_1 and R_2 are not necessarily distinct).
foreign key constraint		A foreign key in a relation is constrained in the values it might take in that every such value must also be a value of the relevant candidate key in the referenced relation. See also referential integrity.
format integration (data integration)		An aspect of data integration concerned mainly with ensuring domain consistency. In most organizations, there are usually many cases where attributes in one system have different formats from the formats of the same, or similar attributes in other systems.
fragmentation		The process of breaking a database table into parts for efficient placement across a distributed database system.
free text storage system		A computer system that stores and retrieves text in the form of strings of characters, punctuation, formatting symbols, etc. without attempting to impose any additional structure on the text they contain.
full outer join		A join which returns a table, in which rows from both joined tables that do not have a matching entry in the other joined table are also included in the final table, with NULL in the missing column entries.

fully functionally dependent		This is a property of attributes. If A and B are sets of attributes of a relation R such that $A \mapsto B$ and B is NOT functionally dependent on any proper subset of A , then B is said to be fully functionally dependent on A . This is equivalent to saying that A is an irreducible determinant. This implies that if B is functionally dependent on a proper subset of A , then B is NOT fully functionally dependent on A .
function		A programming language routine that takes a number of arguments and returns a single value.
functional dependency		An attribute A of a relation R is functionally dependent on a set of attributes $S = \{S_1, \dots, S_n\}$ of R if each value $(s_1, \dots, s_n)$ (where s_1 is a value of S_1 , s_2 a value of S_2 and so on) determines a single value of A . The notation is $S \mapsto A$
galaxy schema		A collection of star schemas. Also known as a fact constellation.
gateway		A standard or proprietary interface between an external information source (server) and an application process (client), which allows the client to generate commands, possibly SQL statements, to be executed at the information source (server) to retrieve the required data. Examples of gateways include Open Database Connectivity (ODBC), Object Linking and Embedding for Databases (OLE-DB), and Java Database Connectivity (JDBC).
global logical schema	see global schema	
global schema		A description of all the data accessible within a DDBMS.
grammatical analysis, grammatical parse		Analysing textual descriptions of an enterprise, in order to develop an entity-relationship model that represents the data requirements of that enterprise by identifying nouns and noun phrases, and verbs and verb phrases.
grouped table		In the Logical Processing Model (see Block 3) the intermediate table that results from the processing of a GROUP BY clause.
grouping columns		The columns specified in an SQL GROUP BY clause.
heading (of a relation)		A heading of a relation consists of a fixed set of attributes $(A_1, A_2, \dots, A_n)$ such that each attribute A_i corresponds to exactly one domain D_i ($i = 1, 2, \dots, n$).
hierarchic model		A type of database model in which the data items form a hierarchy of data in which each data item has at most one parent data item. Data must be located by navigating through the hierarchy. Very efficient for certain kinds of regular processing, but does not adapt well to ad-hoc queries.
hierarchical data structure		An organisation of data in a hierarchy of levels.

HOLAP		Hybrid OLAP – a combination of MOLAP and ROLAP systems
host computer		The computer on which a DBMS or other software system is executing.
host language		A programming language used for an application program that contains embedded SQL.
host variable		A variable of a host (programming) language defined so that it can be used within embedded SQL statements.
identifying attribute		The collection of attributes that can be used to uniquely identify each instance of an entity type. E.g. The identifying attribute for a Car entity type might be the registration number. Sometimes called an identifier.
impedance mismatch		The difference in processing styles between different software components. Most frequently used to when referring to the transfer of data between SQL databases, which use set-at-a-time processing and data representations, and traditional programming environments, which use value-at-a-time processing.
inclusive OR		In two valued logic, if OR is considered inclusive, then the expression A OR B is TRUE if A is TRUE or B is TRUE or if both A and B are TRUE.
inclusive relationships		A pair, or more, of relationships whose occurrences must involve the same entity instances in each relationship occurrence. Recorded in the course data modeling by additional constraint statements.
inconsistent data		Contradictions between data in one part of the database and data in another.
incorrect data		Data that does not correctly reflect the reality that the database is modeling.
index		A storage schema construct that can permit more efficient access to data. It tells you where in the storage schema a specific data item is stored and allows direct access to the data item without the need for sequential searching (compare with a book index).
information		Meaningful data. Data that has been interpreted with reference to some given context.
Information Commissioner		The UK government office responsible for overseeing both the Data Protection Act and the Freedom of Information Act. Those holding and processing personal data must inform the information commissioner's office.
information integration		The process where data that is maintained in two or more information sources is extracted to build a single repository, possibly virtual, containing information from all sources, so the data can be queried as a unit, easily and efficiently.
information requirements		Specification of the requirements of a database. Information requirements have two components – data requirements and operational requirements.

information system		Usually considered to be a computer-based system that supports decision-making, however information systems existed long before computers appeared.
information systems		The discipline concerned with “the organized provision of information in support of people taking purposeful, or intentional, action, this action usually – but not exclusively – initiated within organizations.” Pg 31, Information, Systems and Information Systems, Checkland and Holwell, Wiley international Publ.
initial table		In a logical processing model, the initial tables are the tables that form the first stage of the processing (normally given in the FROM clause of the main query).
inner join		An operation which selects pairs of rows from two tables by matching the contents of two or more related columns.
insert row		A row of a result set that is used for inserting a new row into a table.
insertion anomaly		An anomalous situation in which it is impossible to insert new occurrences of single-valued facts into a relational table already containing occurrences of that single-valued fact.
integration component (of data warehouse architecture)		The component that cleans all the data selected from the different information sources, and combines the data, resolving any incompatibilities between the same types of data from different sources, and brings the data into a consistent format ready for loading into the warehouse database. Comprises the set of integrators for a data warehouse. Each integrator is responsible for installing the data from a particular information sources into the warehouse database.
integrator (module)		The module is responsible for installing the data from the different information sources into the warehouse database
integrity constraints		Constraints that maintain the consistency and correctness of data.
interactive requests		Interaction with a system that involves a real-time response to actions taken by a user of the system.
interface		A description of the data and access mechanisms used to access data held in storage.
intermediate table		In a logical processing model, any table that is generated as a step in the processing after the initial table, but before the final table.
interoperable systems		Two or more systems that can interact by exchanging data or processing instructions.
intersection (relational operator)		Given union-compatible relations A and B , A intersection B is a relation with heading the same as A and B and with body the set of all those tuples which are both in the body of A and in the body of B .

intersection entity type		An entity type that replaces a many-to-many relationship when it is recognised that the relationship has data that is to be recorded. Generated when a many-to-many relationship is resolved.
intersection relation		This is a relation which exists only to represent an m:n relationship between two entity types.
irreducible determinant		The determinant P of the functional dependency $P \mapsto A$ is said to be irreducible if there is no proper subset S of P such that $S \mapsto A$. If such a subset does exist, then P is said to be reducible.
ISO		International Organisation for Standardisation.
isolation level		The degree to which the operations in one transaction can be seen by other transactions. Used to limit the amount of concurrent access that is allowed to a database.
iterator		A Java object that provides the means to fetch rows of a result set.
JAR file		Java Archive file; a file containing Java classes.
Java Server Pages		A means of embedding fragments of Java code (scriptlets) in web pages (translated into servlets for execution).
Java virtual machines (JVM)		A software process that creates an execution environment for programs written in standard Java programming language. Any computer that has a Java virtual machine written to execute on it will execute standard Java programs in exactly the same way. This allows highly portable Java code to be written and distributed easily.
JDBC driver		A software component that provides a JDBC interface to a particular DBMS.
JDBC-ODBC bridge		A JDBC driver that makes use of an ODBC driver to connect to a DBMS.
join (relational operator)		A relational operator which combines two relations into a single relation in the following way: Suppose we have two relations R_1 and R_2 which have a set of attributes $\{A_1, A_2, \dots\}$ in common (in the same way that <i>StudentId</i> is a common attribute of the relations <i>Student</i> and <i>Enrolment</i>). That is, the heading of R_1 is $R_1(X_1, X_2, \dots, A_1, A_2, \dots)$ and the heading of R_2 is $R_2(Y_1, Y_2, \dots, A_1, A_2, \dots)$. None of the attributes $X_1, X_2, \dots$ is equal to any of the attributes $Y_1, Y_2, \dots$ – they have different names and/or different domains. Then R_1 join R_2 is the relation with heading consisting of the set of attributes $\{X_1, X_2, \dots, A_1, A_2, \dots, Y_1, Y_2, \dots\}$ and body consisting of the set of tuples of the form $\langle x_1, x_2, \dots, a_1, a_2, \dots, y_1, y_2, \dots \rangle$, where $\langle x_1, x_2, \dots, a_1, a_2, \dots \rangle$ is a tuple of R_1 and $\langle a_1, a_2, \dots, y_1, y_2, \dots \rangle$ is a tuple of R_2 .

join condition (SQL)		States which columns in two joined tables must contain equal values.
key		A synonym for candidate key
keyword		A fixed set of words built into SQL (such as SELECT , FROM , INSERT , CREATE etc).
Law of Excluded Middle		A theorem of classical logic, which states that the proposition (P OR NOT (P)) is always TRUE, whether P is TRUE or FALSE.
lazy approach (of data warehousing)		When a query is posed, the appropriate set of information sources to answer the query is determined, and the appropriate queries or commands for each information source are generated to extract the data.
leaf node		A node in a tree that does not have a child node.
left outer join		A join which returns a table, in which rows from the first joined table that do not have a matching entry in the second joined table are also included in the final table, with NULL in the missing column entries.
legacy system		A computer or information system that was built using 'old' technologies. Usually represents a system that cannot easily be ported to a new environment and is therefore maintained in its existing form despite problems caused by the use of out-dated software or hardware.
library		A file on a computer system containing a collection of stored programs and procedures accessible to user and system processes.
lifecycle		The sequence of processes involved in the development of a product.
limitations		A statement of the limits under which a conceptual data model will be valid.
local autonomy		Ensuring that all processing and storage of data takes place close to each other. Also known as site autonomy.
location independence (relating to data)		The ability of the user to access all the data from any site without needing to know or refer to its actual location. A desirable feature of a distributed database system.
locking (SQL data)		A mechanism preventing one or more rows in a table from being updated.
logged source (relating to information source)		Information sources that maintain a change (transaction) log that can be inspected, so that changes of interest can be extracted from the log.
logging		Recording the activities performed by a DBMS to give a record of the changes to the data in the databases since the last back-up copy of the data was made. The back-up recovers the data to the state it was in at the time of the back-up was made, while the log is used to recover the data since that back-up until the time of the failure of the system.

logical data independence		In the three schema architecture the separation between the external schema and the conceptual schema that enables changes to the conceptual schema that do not require changes to the external schema.
logical interface		An interface to data that uses logical references to the names of the data items in the storage files – see database approach.
logical operators		A means of combining conditions. SQL implements the logical operators AND, OR and NOT.
logical processing model		A theoretical model of how SQL queries are evaluated.
logical schema		A description of the data that focuses on its logical characteristics. Also known as a conceptual schema. A part of the three schema architecture, there is one logical schema defined for a given database.
logical view		An abstract description of the database that concentrates on the essential properties and characteristics of the data and its associations rather than on issues of presentation or storage.
lossy decomposition	see non-loss decomposition	
lost update		An inconsistency that can arise during concurrent transactions. A user's transaction updates a row and subsequently uses the values in that row, but another user's transaction alters the row in between. The update is lost over the course of the transaction.
maintenance		The making of necessary changes to a computer system to enable its continued functioning. This can include bug fixes, re-programming to meet changing requirements, or enhancing existing features of the software.
mandatory		The participation condition property of a relationship that indicates that every instance of the entity type is involved in the relationship. Mandatory participation represents a constraint on the values permitted for the entity instances.
many-to-many		The degree representation of a relationship between two entity types A and B that has the reading: Each A is related to one or more Bs. Each B is related to one or more As.
mapping		The translation between two different descriptions of data, for example mapping a logical representation of the names of a data item to the physical representation of the data item in a file store.
mapping schema		A specification of the transformation between classes and tables required for object-relational mapping.
mapping tool		A software component supporting object-relational mapping.

markup language		A language that uses tagging of parts of a document to indicate semantic and structural content that may be used in processing and displaying the document. For example HTML, XML.
markup tag		Notation used in a markup language, often referred to as markup.
materialised view (of the data)		Pre-computed aggregated data in an OLAP system. These views are normally stored in the multidimensional parts of HOPLAP and MOLAP systems.
measures (of multi dimensional data model)		The numerical size or quantity of a fact in terms of specified units.
mediated approach		See lazy approach (of data warehousing).
mediator (module)		In the Lazy approach to data warehousing, the module that decomposes queries and combines the results
metadata		Data about data, usually describing the structure, content or use of the data. Commonly held in a system catalogue or data dictionary.
MOLAP		Multi-dimensional OLAP – OLAP systems that employ a multidimensional database system where data is conceptually stored in cells of a multidimensional array.
monitor		The component is responsible for automatically detecting changes of interest to the data warehouse in the source data, and propagating those changes to the integrator.
monolithic systems		A single large computer system, or software program. Not internally differentiated into objects, clients and servers, or other sub-components.
multi dimensional data model		A conceptual data model that enables decision makers to view data from different, and multiple, perspectives.
multimedia		A term that encompasses a range of types of data that include text documents, audio, still images and moving images.
multi-user system		A computer or information system that allows several users to simultaneously share the systems features and capabilities.
multi-value fact (of an entity type or relationship)		A statement (fact) identifying a property of an entity type or relationship which can take several values for each occurrence of that entity type.
named iterator		An iterator that is used to assign values to host variables by methods having the names of columns.
native interface		The means by which an application program interacts with a DBMS that is a designed feature of that DBMS.
natural join		An inner join that matches on all the columns with the same name in the joined tables.

natural language predicate		A statement, expressing in a natural language such as everyday English, which when applied to a particular subject, has a truth value. That is, the statement is either true or false.
network model		A type of database model in which the data items form a network of data, that is like the hierarchical model but where each data item can have more than one parent data item. (Compare with hierarchic model).
node (of a tree)		A component connected to other nodes within a tree.
non-loss decomposition		If the project operator is applied to a relation to give two relations with overlapping attributes, and then these two are recombined using join , then if the original relation is restored, the original project decomposition is called a non-loss decomposition. If this doesn't happen, the resulting relation has greater cardinality than the original, and the decomposition is said to be lossy.
non-repeatable read		An inconsistency that can arise during concurrent transactions. A row is read twice in the course of a transaction, but another user's transaction alters the row in between. The first transaction receives different values on each read.
Non-trivial functional dependency	see trivial functional dependency	
normalised tables		Tables that do not include any redundant duplication, which may result in insertion, amendment and deletion anomalies.
not null constraint		An SQL column constraint that prevents values in that column being NULL.
NULL		A marker indicating the absence of a value in a row
null value		An occurrence of NULL in a row (although strictly, NULL is a marker, not a value of the attribute).
object-oriented		A representation of data that depends on combining a structured data type with the procedures (methods) that access it. (A sophisticated form of abstract data type).
object-relational mapping		Enabling a Java program to access a relational database by using Java classes that are mapped to tables.
occurrence (of a single valued fact)		This is a particular instantiation of a single-valued fact.
ODBC	see Open Database Connectivity	
ODBC data source		Defined use of an ODBC driver with a specified database

ODBC driver		A software component for a database that converts ODBC procedure calls into the procedures calls of a proprietary DBMS interface. Thus allowing a proprietary database system to be accessed by a user application written to use the ODBC interface.
OLTP	see On-Line Transaction Processing	
one-to-many		The degree representation of a relationship between two entity types A and B that has the readings: Each A is related to one or more Bs. Each B is related to at most one A.
one-to-one		The degree representation of a relationship between two entity types A and B that has the readings: Each A is related to at most one B. Each B is related to at most one A.
On-Line Transaction Processing (OLTP)		The type of processing of data that expects a user to interact directly with the database in real-time, usually involving a series of queries (a transactions) that must be co-coordinated as a single unit of work. Typical examples include on-line ordering or booking systems.
Open Database Connectivity (ODBC)		An open (standard) interface application programming interface to a DBMS offering general procedures for data access and available for common use. It has a standard specification known as SQL Call-Level Interface (SQL/CLI).
operand		An 'input' to an operator; an object on which the operator acts.
operating system		The software that directly controls the underlying hardware and other software in a computer system. For example: Windows, Unix, Linux, DOS, etc. are types of operating system.
operational maintenance		Ensuring that the database continues to satisfy the information requirements after it has been installed by monitoring its performance and undertaking a database restructuring or database reorganization when necessary.
operational requirements		The part of the information requirements that describe how the data, and the relationships between the data, needs to be processed.
optimizer		A component of a relational DBMS which is concerned with efficiency. (see also query optimizer)
optional		The participation condition property of a relationship that indicates that not all instances of the entity type are involved in the relationship.
origination costs		The specific costs associated with acquiring data and making it fit for purpose within an information system.

outer join	see also left outer join, right outer join and full outer join.	A join in which unmatched rows are replaced with NULL.
outer query		The query in which a subquery is embedded.
outer reference		A reference in a subquery to a column contained in an outer query.
owner (of a table)		The user who created a table, or who has been granted ownership of a table by the creator of the table. Owners have certain privileges on the tables that they own that other users do not.
parameter		A combination of place holder and value passed to the code of a procedure, so that the procedure can access the value and use it within the procedure code.
parameter marker		The symbol ? placed in a character string (an SQL statement) given as an argument for an invocation of the <code>prepareStatement</code> method, which is assigned a value by a set method before execution of the SQL statement.
parent node		A node that is connected to one or more child nodes.
participation condition		A property of a relationship that indicates how two or more entity occurrences are related to each other. Participation conditions of interest are optional and mandatory.
pattern (SQL LIKE)		A string representing the <i>set</i> of strings that can be matched using LIKE .
PDAs		Personal Digital Assistants, small but powerful hand-held computers often with mobile database facilities.
persistent data		Data that remains stored and accessible after the processes manipulating the data have terminated.
persistent programming language		A programming language that maintains the state of its variables and data structures between executions of the program.
phantom insert		The insertion of phantom rows into a database.
phantom row		An inconsistency that can arise during concurrent transactions. A condition on multiple rows is evaluated twice in the course of a transaction, but another user's transaction alters the table in between. In the first transaction, the condition fails the first time it is evaluated, but succeeds the second. The phantom rows result in the condition evaluating differently.
physical data independence		In the three schema architecture the separation between the conceptual schema and the internal schema that enables changes to the internal schema without the need to change the conceptual schema.
physical data warehouse		A data warehouse where the data is stored separately from the data sources. Compare with virtual data warehouse.

physical implementation		The realization of a software design on specific hardware and specific software. Usually indicating that specific decisions about how to represent data and programming logic have been determined.
physical interface		An interface to data that makes visible the physical structure of the data in the storage files – see file-based storage.
physical view		The description of the database that concentrates on the physical storage representation of the data on the host computer.
porting		Maintenance performed on a user process to allow that process to be used in another operating system environment or on different hardware.
positioned iterator		An iterator that is used to assign values to host variables according to their position in a fetch statement.
positioned updates		An update carried out at the current cursor position.
posted foreign key		An attribute which is introduced into a relation only to represent a relationship.
post-relational		A description of database representations that were developed after the relational representation – usually used to refer to Object, Object-Relational and XML databases.
pre-compiler		A process that converts embedded SQL in the source code of an application program to statements in the host programming language, giving source code that contains just the host programming language.
predefined data types		Data types that are built into SQL.
preposted foreign key		A foreign key which matches an attribute(s) already present in the corresponding entity. See weak entity type.
primary key		A particular key chosen from the set of candidate keys of a relation.
privacy		The right of an individual to control the use of personal information, including financial, health and lifestyle data.
privileges		The actions that specified users may perform on a table (such as querying, updating etc).
problem statement		A question that a user wishes to use the database to answer, that does not refer to the specific tables or columns in the database.
procedure		A programming language routine with multiple parameters that does not explicitly return a value.
procedure call		A mechanism in a programming language used to active a section of program code (known as a procedure) and on completion of that code to return control to the point after the procedure call was made. This permits commonly performed tasks to be declared once and used whenever required.

process	see user process	Usually an abbreviated form of user process.
Processing Instruction (PI)		Markup in a document prolog that specifies how an XML document may be processed.
productive maintenance		Maintenance updates limited to those changes specifically required to meet the maintenance needs.
program	see user process	Usually an abbreviated form of user program.
project (relational operator)		The project operator may be thought of as slicing a relation vertically, picking out specified attributes. The syntax is project <relation> over <attribute list>
proprietary		A feature or capability that is owned by a specific vendor. Commonly used to refer to interfaces that are specific to a user's product, or to non-standard extensions that are available only one vendor's products.
pull systems		A replication system in which a database site requests an update to the data items it holds that have been replicated from other databases.
push systems		A replication system in which changes to a data item are pushed to the database containing replicas of the changed data items.
qualified names		The name of a column prefixed by the name of the table in which the column appears.
quality management		The process of establishing standards of quality for a product or service, establishing procedures or methods that ensure that these standards of quality are met, monitoring the actual quality and taking control when quality falls below acceptable thresholds.
quantified		An expression containing a quantifier is said to be quantified.
quantifier		Operators that specify how many elements in a subquery must satisfy a given condition (ALL, ANY).
query (general)		Requests entered into a DBMS to access or manipulate data.
query (SQL)		Short form of 'query statement'. A query always returns a table.
query language	see data manipulation language	
query optimizer		A component of a DBMS that processes a user query in order to produce a more efficient internal query that answers the same request as the user query. An essential component of relational database systems (including SQL-base systems) where the query is expressed in logical terms, without regard for the internal physical storage. Most query optimizers make use of formal relational theory as well as pragmatic rules of thumb in translating between query representations.

query specification		A query that contains one occurrence (at most) each of the keywords SELECT , FROM , WHERE , GROUP BY and HAVING .
query statement		A query which contains one or more query specifications (every query specification is a query statement).
queryable source (relating to information source)		Information sources that allow the wrapper/monitors to query the data at the information source, so that periodic polling can be used to detect changes of interest.
recovery		Following the failure of a computer system the actions required to restart and rebuild the system to the point where it is useable.
recursive relationship		A relationship which involves the same entity type in different roles in the relationship.
reducible determinant	see irreducible determinant	A is an irreducible determinant of the functional dependency $A \rightarrow B$ if there is no proper subset C of A such that $C \rightarrow B$ is a functional dependency.
redundant duplication		The occurrence of the same data in more than one place in a database in such a way that, should one of the recorded values be lost it is possible to recover that data from the remaining data in the database. Data that can potentially lead to consistency problems without appropriate management.
referenced relation		A relation from which a foreign key is derived, that is, which has a candidate key such that all the values of a foreign key in the referencing relation are also values of the candidate key.
referenced table		The table containing the column(s) referenced by the foreign key contained in a referencing table.
referencing relation		A relation with a foreign key.
referencing table		A table containing a foreign key.
referential actions		Actions taken by the database when a row being deleted from a referenced table has dependent rows. The actions describe changes that should be made to the dependent rows so that the changes do not violate the referential integrity of the database.
referential constraint (SQL)		Requires that each non-NULL value in the foreign key of a referencing table has a matching value in the referenced table. Equivalent to the foreign key constraint in relational theory.
referential integrity (relational model)		A rule which maintains the integrity of a foreign key. If a relation, R_2 , has a foreign key, F , that references a candidate key P in a relation R_1 , then every value of F appearing in a tuple of R_2 must equal a value of P appearing in a tuple of R_1 . Note that P is usually the primary key of R_1 , and that R_1 and R_2 need not be different.

referential integrity rule (SQL)		If a table, T_2 , has a foreign key, F , that references a candidate key, C , in another table, T_1 , then every value in T_2 . F must either be equal to a value in $R_1.C$ or be NULL.
relation (formal)		(i) A relation R on domains $D_1, D_2, \dots, D_n$, not necessarily distinct, consists of a heading and a body. (ii) (With respect to predicate logic) A relation is a set of all the known true propositions that can be instantiated from a given natural language predicate.
relation (informal)		The representation of data in the relational model. A relation is similar to, but has more restrictive properties than, an SQL table.
relational database		A database whose logical interface follows the relational approach. Compare with relational model and SQL database.
relational model		A mathematical representation of data built on sound formal principles, and that structures data in relations (table-like) structures.
relational operator		A relational operator is a function which acts on (is applied to) relations (its operands) to yield other relations.
relational table		A table representing a relation. Such a table must obey the 5 rules described in Block 2, Section 2.1.
relationship		A link or association between entity types.
relationship occurrence		A specific link or association between two entity occurrences.
rename (relational operator)		An operator which has the effect of taking a relation as operand and yielding a relation identical to the first except that the names of specified attributes have been changed. Syntax: <i>R rename (A1 as NewA1, A2 as NewA2, ...)</i>
replication		Data is copied to other database systems and managed so as to give the appearance of only a single copy of the data existing. I.e. consistency problems are managed by a DDBMS. Data is copied so that all processing at a site can be performed with data held at that site, thus making it efficient.
replication server		A server process responsible for managing replicated data, ensuring consistency of data items when updates occur.
request	see problem statement	
resolving a many-to-many relationship		The process of replacing a m:n relationship with an intersection entity type and appropriate 1:m relationships.

restricted effect		A means of ensuring referential integrity by not allowing referenced tuples to be deleted.
result set		An ordered collection of rows resulting from execution of a query, which may be referenced by a cursor.
right outer join		A join which returns a table, in which rows from the second joined table that do not have a matching entry in the first joined table are also included in the final table, with NULL in the missing column entries.
ROLAP		Relational OLAP – OLAP systems that employ a relational database system to store data as <i>fact tables</i> and <i>dimension tables</i> .
role		A set of authorisation identifiers with which users can be labeled so that a group of users with the same label can be granted privileges on database objects together.
roll up (relating to multi dimensional data model)		An operation performs an aggregation on a multidimensional view of data, either by ascending a concept hierarchy for a dimension, or by reducing the number of dimensions. The result of operation is to produce summary data from detailed data along one or more dimensions. Also known as drill up.
rollback		Undoing the work that has been done by a transaction before a commit.
root node		A node in a tree data structure that does not have a parent node.
routine		A series of statements stored by the DBMS and executed as a single process.
row		A set of data values which represents a single item in a table. A row contains one value for each column in the table. Similar to a tuple in relational theory.
row number		The position of a row in a Java result set.
rowset		A set of rows that can be obtained by a Java fetch operation.
savepoint		A marker in a transaction which allows all the work up to that savepoint to kept, and only the work that came after the savepoint to be rolled back.
schema		A description of data, its properties and characteristics, that is capable of being processed by a DBMS.
scripting language		A programming language that is usually high-level, dynamic, interpreted and can often be executed interactively.
scriptlet		A fragment of Java code embedded in a web page.
scrollable cursor		A cursor defined as having the ability to scroll to any row within a Java result set.
search condition		The condition in the WHERE or HAVING clause of a query.

Second normal form		A relation is in second normal form (2NF) if and only if every non-primary key attribute is fully functionally dependent on the primary key.
second-cut design		Any design produced by flexing a first-cut design.
security access restriction		A DBMS permission that permits certain users to perform certain processing on certain data.
select (operator)		select is a relational operator which, when applied to a relation, yields those tuples in the relation which satisfy a certain Boolean condition. select may be thought of as splitting a relational table horizontally. The syntax is: select <relation> where <selection condition>
selection condition		A Boolean expression which is applied to each tuple of a relation in turn to identify those which are returned by the select operator.
self joins		A join that joins two instances of the same table.
semantic feature		A feature or property of data that is related to the meaning, not just the structure or representation of the data.
semantic integration (data integration)		An aspect of data integration concerned with the consistent meaning of data extracted from different operational systems.
sequential access		Data is read in an ordered sequence, usually first item, second item, ...etc. This contrasts with direct access.
serialisation (of a tree)		The conversion of the nodes of a tree into a sequence.
serializable		A set of concurrent transactions can be executed in a way equivalent to a serialized execution.
serializable isolation		Isolation level that guarantees serializable execution of a transaction.
serialized execution		Transactions can only be carried out one at a time. Each transaction must complete before the next is executed.
server		The processes in a client-server system that receives requests for services it offers, satisfies the request and returns the result to the client that request it.
servlet		A Java component that provides content for web pages to a web server.
shared data		Data that can be used by more than one process; sharing is usually limited by access control and concurrency management.
shredding		The process of decomposing an XML document into simple values that can be stored in one or more database tables.
simple queries		A query that contains no nested queries (also a “query specification”).

single-user system		A computer or information system that permits only one user at a time to make use of the systems features and capabilities. Note that a multi-user computer could host a single user information system.
single-valued fact		This is a statement (fact) identifying a property of an entity type which can only take a single value for each entity.
single-valued fact type		A synonym for single-valued fact
site autonomy	see local autonomy	
snapshot		A copy of the data in a table (or the entire database) either for back-up purposes, or to make certain processing more efficient. A snapshot represents data in the table at the instant the snapshot was made, it is not kept up-to-date if the table it copied is updated.
snapshot source (relating to information source)		Information sources that do not provide triggers, or change logs, and are not queryable. Instead, periodic dumps (snapshots) of the data at the information source must be performed, and changes of interest detected by the comparison of successive snapshots.
snowflake schema		A variant of the star schema represents concept hierarchies using several dimension tables, one table for each level of the concept hierarchy.
software engineering		The application of engineering principles to the development of software artifacts.
software engineering artifacts		The documents and other intermediate products produced during the software engineering activities.
SQL		Structured Query Language.
SQL Anywhere		An implementation of an SQL database engine developed by Sybase, Inc.
SQL database		A database whose logical interface follows the SQL standard. Data in an SQL database is structured as tables that are similar to but not identical to mathematical relations.
SQL implementation		A particular version of SQL released by a software vendor (for example, Oracle SQL, SQL Anywhere etc). An implementation typically contains only a subset of the Standard, and several vendor extensions.
SQL:1987, SQL:1989, SQL:1992, SQL:1999, SQL:2003		Different versions of the Standard, along with the year of their publication.
SQL2		Another name for SQL:1992; published by ISO as ISO 9075:1992.
SQLJ		Embedded SQL for Java.

Standards body		A group of people and businesses who are responsible for maintaining and updating a software standard, such as the SQL language standards.
star schema		A data warehouse database design where an n -dimensional view is represented by a fact table and n -dimension tables.
start-tag		A label denoting the beginning of an element, which may include attributes of the element.
statement of requirements		A document containing descriptions of the required data items, their associations and properties, to be included in a database. A summarised form of the domain of discourse – often omitting “common” knowledge.
static constraints		Constraints on the data that may appear in a database that hold at a particular moment. Contrast this with dynamic constraints.
static SQL		SQL statements that are compiled in an application program and cannot be changed when executed.
storage schema		A description of the data that focuses on the physical storage of the data in the host computer system. Also known as an internal schema. A part of the three schema architecture, there is one storage schema defined for a given database.
stored procedure		A routine defined in SQL with multiple parameters that does not explicitly return a value.
strong entity type		An entity type whose existence does not depend on the existence of another entity types occurrences.
structured data		Data that has a regular, defined, structure that supports the consistent interpretation of the data.
Structured Query Language (SQL)		The ANSI standard database language (sometimes pronounced SEQUEL, but more often as S-Q-L) which developed from the System R language at IBM in the mid-70's. A standard database language utilised by most vendors. However, most vendor products include extensions and variations to the Standard.
stylesheet		A specification of how an XML document should be transformed or displayed.
subject table		The table on which a particular trigger is defined.
subquery		A query specification that occurs in the FROM , WHERE or HAVING clause of another query (the outer query).
SVF	see single-valued fact	
system catalog		The internal description of the data held in a database by a DBMS. Sometimes known as the DBMS data dictionary.
system tables	see DBMS system schema	

systems costs		Any cost involved in the acquisition and implementation of a new information system.
table		A set of data values, organised into rows and columns. A table is the SQL equivalent of a relation in relational theory although with fewer structural limitations.
table constraint		A constraint defined on a table.
target table		A table containing the data being changed or queried by a data manipulation statement.
temporal properties		Multi-valued properties of entity types and relationships where each value is dependent on time.
thin-client		A client process that is responsible only for the display and entry of data and undertakes none of the business application logic for an application.
Third normal form		A relation is in third normal form (3NF) if and only if it is in 2NF and no non-primary-key attribute is transitively dependent on the primary key.
three-schema architecture		An abstract description of the architecture of a database system that represents the data at three distinct schema levels – external (user), logical (conceptual) and storage (internal). Also known as the ANSI/SPARC three-level architecture.
times (relational operator)		Given two relations A and B such that no attribute in A is the same as one in B , A times B is the relation whose heading is the set union of the attributes of A and B , and whose body is the Cartesian product of the bodies of A and B .
transaction		A collection of requests or queries that should be considered to be a single unit of work and treated as a single activity. E.g. transferring funds from one bank account is a transaction that involves two queries both of which must complete for the transfer to be successful.
transitive dependence		An attribute A is transitively dependent on a set of attributes X in a relation R if there is a set of attributes Y such that all the following properties hold: TD(i) $X \twoheadrightarrow Y$ and $Y \twoheadrightarrow A$ TD(ii) It is not true that $Y \twoheadrightarrow X$ TD(iii) A is not an attribute of either X or Y .
transitivity of functional dependencies		This is a property of functional dependencies that if $S \twoheadrightarrow A$ and $A \twoheadrightarrow B$ then $S \twoheadrightarrow B$.
traversing (a tree)		Moving from one node to another to visit all nodes in a sequence.
tree		A hierarchical data structure formed from connected nodes.
trigger		A procedure that is automatically executed when its subject table is updated.

trivial functional dependency		A trivial functional dependency is one in which the right hand side is a subset of the determinant, that is, it is of the form $A, B \mapsto A$ or $A \mapsto A$. A non-trivial functional dependency is a functional dependency which isn't trivial.
tuple		A tuple of relation is a set of values of the form $\{A_1: v_1, A_2: v_2, \dots, A_n: v_n\}$, such that each v_i is a value from the domain D_i corresponding to the attribute A_i ($i = 1, 2, \dots, n$).
tuple constraint		A constraint in which Boolean conditions are imposed on values of attributes in the same tuple for each tuple of the relation.
two-phase commit		A DDBMS requires a more complex commit operation in order to co-ordinate the distributed commit instructions at multiple sites. The two-phase commit can be used for this purpose as it is a distributed transaction co-ordination protocol. All participants signal their readiness for commit or rollback to a coordinator node, which then orders the participants to either commit or rollback.
two-valued logic		Logics in which a proposition is either TRUE or FALSE.
unary operator		An operator which acts on (is applied to) just one operand at a time.
uncommitted data		Data that has been inserted into a database as part of a transaction, but which has not been made permanent by the use of commit.
underlying data type		A predefined data type in SQL that is used to define a domain.
underlying tables		The base tables containing the source data for a view.
union (a relational operator)		Given union-compatible relations A and B, A union B is a relation with heading the same as A and B and body the set of all those tuples which are either in the body of A or in the body of B or both.
union compatible (SQL)		Two tables are union compatible if they have the same number of columns, and corresponding columns must be of the same predefined data type.
union-compatible (relational)		Two relations are said to be union-compatible if they have the same set of attributes, that is, if every attribute in one has the same name and domain as an attribute in the other.
unique constraint		Requires that no two rows contain the same value in a particular column (or that no two rows contain the same combination of values in a set of columns).
UNKNOWN		A third truth value used to extend two-valued logic, so that a proposition can be TRUE, FALSE or UNKNOWN.

unnormalized tables		Tables that may include redundant duplication, which may result in insertion, amendment and deletion anomalies.
unproductive maintenance		Changes made to specific components of a system for maintenance purposes that result in the need to make additional changes to indirectly related components.
updatable		A view is updatable if rows can be inserted into the data defined by the view or can be deleted from it, or if values in rows can be changed. This requires that the DBMS be able to establish a mapping from the view to the underlying base tables.
user application	see user process	
user process		Software that makes use of a database – user process (or program or application) is used when the term refers to an application or program running on behalf of an end-user, rather than an operating systems process, DBMS process or system process.
user program	see user process	
user tables		The tables in a database (managed by the DBMS) that are holding data for user processes. Compare with system tables.
user view		The data and its specific presentation (appearance) displayed to a user by an application process. The application process determines how to present the data to the user; the data accessible by the application process is determined by the external schema the application process uses.
user-defined data type (SQL)		A data type that is defined by the user, as opposed to a predefined data type.
user-defined function	see function	
valid XML		A text document with XML markup that conforms to a document definition.
value expression		An expression that carries out operations (such as addition, or concatenation) on the values in one or more columns of tables.
value set		The set of permissible data values that the attribute of an entity type may hold.
view (SQL)		An SQL construct that permits a table of data to be described by a query using other views or base tables. The query is evaluated to generate the data whenever data in the view is required.
view maintenance		The process of updating materialized views in response to changes to the underlying unaggregated data.
virtual data warehouse		A data warehouse where the data is stored separately from the data sources. Compare with physical data warehouse.

Waterfall model		A (notionally) linear series of software development activities that starts with requirements analysis and finishes with implementation and maintenance.
weak entity types		A weak entity type is one whose occurrences depend on the existence of another entity occurrence. Usually recognisable because a weak entity type identifier will be a combination of identifying attributes from the other (strong) entity type. E.g. an enrolment of a student on a course is dependent on the existence of a student and a course. (Note weak entity types can be hidden when surrogate identifiers are freely used in data modeling).
Web servers		A server that offers web services and is accessible via internet or intranet facilities.
well formed query		A syntactically correct SQL query, irrespective of whether it correctly answers the user's question.
well-formed XML		A text document with XML markup conforming to the XML specification.
wrapper (module)		In the eager approach to data warehousing, a module that is responsible for translating data from the native format of the information source into the format and data model used by the warehouse database.
wrapper generator		Software that creates a wrapper which can be used to automate (or semi-automate) the process of implementing wrapper/monitor modules.
XML		Extensible Markup Language; a markup language specified by the World Wide Web Consortium (W3C). Frequently used for the representation of data in document form for communication, storage and presentation.
XML application		A program that processes XML.
XML parser		A program that analyses an XML document to give the structure of its elements.
XML schema		A formal description of the structure of an XML document and its data types expressed according to the XML schema specification.
XML value (for SQL)		An instance of the SQL data type XML.
XPath		A means of selecting elements in an XML document by specifying a path.
XQuery		A means of querying an XML document.
XSL-FO		XML Stylesheet Language Formatting Objects, the notation that can be used to specify how one kind of XML document can be transformed into another kind of formatted document.
XSLT		XML Stylesheet Language Transformation, the notation that can be used to specify how one kind of XML document can be transformed into another kind of XML document.